

Group 5
Advanced NLP

Summary of the Paper:

Mikolov, Tomas et al.

"Distributed Representations of Words and Phrases and their Compositionality"

Group Members:

Raphael Steinborn | 351421

Lidia Makishti | 346621

Vineeth Racharla | 349073

Saqib Sarwar | 351757

1 Introduction – Vineeth Racharla

In the paper “Distributed Representations of Words and Phrases and Their Compositionality,” the researchers offer an improved skip-gram model for learning word vectors, which greatly advances the quality and speed of word representations. Key milestones consist of subsampling for frequent words to boost training as well as representation of infrequent words; it also introduces negative sampling as an alternative to hierarchical softmax. These techniques not only make training more efficient but also help the model capture better syntactic and semantic relations between words. Besides, another issue that is addressed in the paper is representing phrases and idiomatic expressions used in detecting such phrases and forming vector representations.

Objective

The outcome of this research paper is to increase the capabilities of Skip-gram model to further enhance the word vector quality, accelerate the training process, and effectively handle the representation of phrases that cannot be simply created from their individual word components.

To achieve these goals, the researchers suggested a number of significant changes and enhancements to the existing model of Skip-gram. During the training phase, they first incorporate frequent word subsampling. By decreasing the number of regular words that must be processed frequently, this method greatly expedites the training process while simultaneously enhancing the quality of the representations for uncommon words. The model can catch more significant links between words and improve the overall accuracy of the word vectors by concentrating on less common terms.

In addition to the hierarchical SoftMax strategy, the study proposes a negative sampling method for training the skip-gram model. By using fewer negative samples than positive samples, negative sampling reduces training times and increases representation accuracy, especially for terms that appear frequently. This solution is also more computationally efficient because it does not involve computing the entire SoftMax function. The paper, for example, shows that the hierarchical softmax approach with Huffman codes required 41 minutes to achieve a total accuracy of 47% on the analogical reasoning task, whereas negative sampling with five negative samples required 38 minutes of training and a total accuracy of 59%.

Another important contribution of this research paper is the development of a process for representing phrases as unique tokens rather than just combinations of their constituent words. This process mainly addresses the drawbacks of initial word vector representation, which have difficulty in understanding multi-word and idiomatic sentences. For example, the meanings of "New York" and "Times" cannot be easily combined to represent "New York Times," a widely known newspaper. During the training process, the model can create more accurate and intelligible vector representations for phrases like "New York Times" by treating them as separate units, hence increasing the model's ability to handle more complex language structures.

In Conclusion, the main objective of this paper was to improve the Skip-gram model's word vector quality, speed up training, and enable effective phrase representation. Natural language processing (NLP) will benefit greatly from these enhancements, which will finally increase the model's effectiveness and capacity to capture a greater range of linguistic complexities.

2 Importance of Contribution – Lidia Makishti

This paper presents significant contributions to both research and practice in the field of NLP with a focus on the Skip-gram model. Some key aspects that we will elaborate on this document are the model's efficiency, quality of vector representation, challenges in handling phrases and practical applications. All these important factors make up a valuable part of the continuous Skip-gram model for both academic research and industry applications, highlighting its importance and influence in the field of NLP.

Efficiency

In traditional neural network architecture, you will find dense matrix multiplications for learning word vectors, but on the contrary the Skip-gram model's training speed doesn't require any expensive computational process. This makes the training of the model extremely efficient: an optimized single-machine implementation can train on more than 100 billion words per day. Additionally, sub-sampling of frequent words during training results in a significant speedup, around 2x-10x, and improves the accuracy of representations of less frequent words. These representations capture intricate syntactic and semantic relationships, leading to an improved performance.

Quality of Representations

The paper enhances the quality of word vector representations through different training methods. The first one was hierarchical softmax, which uses a binary tree representation of the output layer and for each node, it represents the probabilities of its child nodes. This kind of tree structure that is used by hierarchical softmax has a large effect on the performance. As an alternative to it, there is Negative Contrastive Estimation (NCE), which suggests that a good model should be able to differentiate data from noise by means of logistic regression. The Skip-gram model is only concerned with learning high-quality vector representations, as a result it simplifies NCE to ensure that the vector representations retain their quality.

Another training method is Negative Sampling (NEG) that replaces every log probability term in Skip-gram objective with a negative sampling objective. NEG task is to distinguish the target word from draws from the noise distribution using also the logistic regression, where there are k negative samples for each data sample. The main difference between these two-training methods is that NCE needs both samples and numerical probabilities of the noise distribution, while Negative Sampling uses only samples.

Furthermore, to counter the imbalance between the rare and frequent words, the paper introduces a simple subsampling approach, that is designed to aggressively subsample words with frequencies greater than the chosen threshold, while maintaining the relative ranking of word frequencies. From training the Skip-gram models the paper comes to the conclusion that Negative Sampling outperforms the hierarchical softmax on the analogical reasoning task and has even slightly better performance than the Noise Contrastive Estimation. Meanwhile, the subsampling of the frequent words improves the training speed several times and makes the word representations significantly more accurate.

Handling Phrases

Many phrases have a meaning that is not a simple composition of the meanings of its individual words. For this issue, the paper suggests a method to learn vector representation for phrases by

identifying words that frequently appear together and infrequently in other contexts. As in “New York Times” will be replaced by unique tokens during training, while a bigram like “this is” will remain unchanged. Therefore, by incorporating phrases into the training process and treating them as unique entities, the Skip-gram model can possibly improve its ability to capture the complex semantics of multi-word expressions and enhance its performance on tasks that involve phrases with non-compositional meanings.

Practical Application

The paper has evaluated the Skip-gram model on other word vectors from the web, and the Skip-gram model has achieved the best performance with a huge margin. The paper continues to show that the big Skip-gram model trained on large corpus visibly outperforms all the other models in the quality of the learned representations. This can be attributed in part to the fact that this model has been trained on about 30 billion words, which is about two to three orders of magnitude more data than the typical size used in the prior work. Interestingly, although the training set is much larger, the training time of the Skip-gram model is just a fraction of the time complexity required by the previous model architectures.

3 Methodology and Procedure – Raphael Steinborn

3.1 Procedure and Algorithms Used

The paper introduces several improvements to the Skip-gram model, enhancing the quality of word vectors and making the training process more efficient. The key procedures and algorithms used can be explained as follows:

The Skip-Gram Model

The Skip-gram model is the core algorithm used for learning word representations. The model's objective is to predict the context words surrounding a given target word. This is done by maximizing the average log probability of context words given the target word across a large corpus of text. Formally, the goal is to maximize:

$$\frac{1}{T} \sum_{t=1}^T \sum_{-c \leq j \leq c, j \neq 0} \log p(w_{t+j} | w_t)$$

Where w_t is the target word, w_{t+j} are the context words within a window size c , and T is the total number of words in the training corpus.

Hierarchical Softmax

To handle large vocabularies efficiently, the authors use hierarchical softmax, which is a method that reduces the computational complexity of calculating the softmax function. This method represents the output layer as a binary tree, where each word in the vocabulary is a leaf node. Therefore, the probability of a word w is determined by the path from the root to the leaf, significantly reducing the number of computations needed.

Negative Sampling

Negative sampling is introduced as an alternative to hierarchical softmax, further simplifying the training process. Instead of calculating the probability for all words in the vocabulary, negative sampling updates a small number of negative samples for each positive example. The objective for negative sampling is:

$$\log \sigma(v_{w_o}^\top v_{w_l}) + \sum_{i=1}^k E_{w_i \sim P_n(w)} [\log \sigma(-v_{w_i}^\top v_{w_l})]$$

Where σ is the sigmoid function, v and v' the input and output vector representations, w_o is the target word, w_l is the context word, and $P_n(w)$ is the noise distribution. This approach significantly reduces computational cost while maintaining high-quality vector representations.

Subsampling of Frequent Words

Frequent words in the corpus, such as "the" and "in", provide less informative context for learning word representations. To address this, the authors use a subsampling approach that probabilistically discards frequent words during training. The probability of discarding a word w_i is given by:

$$P(w_i) = 1 - \sqrt{\frac{t}{f(w_i)}}$$

Where t is a threshold (typically around 10^{-5}) and $f(w_i)$ is the frequency of the word w_i . This technique speeds up training and improves the accuracy of representations for rare words.

Phrase Detection

The authors extend the Skip-gram model to handle phrases, which often have meanings that are not just the sum of their parts. A data-driven approach is used to detect phrases in the text by identifying word pairs that frequently appear together but infrequently apart. This is done using the scoring function:

$$\text{score}(w_i, w_j) = \frac{\text{count}(w_i w_j) - \delta}{\text{count}(w_i) \times \text{count}(w_j)}$$

where δ is a discounting coefficient. Phrases with scores above a certain threshold are treated as single tokens during training, allowing the model to learn meaningful phrase representations.

3.2 Input and Output Data

Input Data

The used training data consists of a large text corpus, specifically an internal Google dataset of news articles with about one billion words. Words occurring less than five times in the corpus were filtered out, resulting in a vocabulary of size 692K.

Output Data

The output of the Skip-gram model is a set of high-dimensional vectors, where each word (or phrase) in the vocabulary is represented as a point in a continuous vector space. These vectors capture syntactic and semantic relationships between words. For instance, vector arithmetic can solve analogies like $\text{vec}(\text{"Berlin"}) - \text{vec}(\text{"Germany"}) + \text{vec}(\text{"France"}) \approx \text{vec}(\text{"Paris"})$.

3.3 Training Procedure

The training procedure can be outlined as shown in Table 1.

Step Number	Step Name	Description
1a	Preprocessing	Tokenize the text corpus and convert words to indices based on the vocabulary.
1b (optional)	Phrase Learning	Identify common phrases in the text corpus and treat them as single tokens.
2	Training	Train the Skip-gram model with negative sampling to learn word vectors. Each word in the corpus is used as a target word, and the model is trained to predict the context words within a specified window size. Negative sampling is applied to select a few negative samples for each target-context pair.
3	Optimization	Use stochastic gradient descent (SGD) to optimize model parameters. The learning rate and hyperparameters are tuned to ensure efficient convergence.
4	Evaluation	Evaluate the quality of vectors using analogical reasoning tasks and benchmark datasets.

Table 1: Training Procedure

4 Results and Conclusions – Saqib Sarwar

4.1 Results

The Skip-gram model along with the suggested improvements excelled in tasks related to reasoning. This demonstrated the model's capability to effectively grasp relationships between words. Mikolov and the team introduced a dataset containing analogies that could be solved by identifying the vector. In this test, the Skip Gram model surpassed both SoftMax and Noise Contrastive Estimation methods thanks to Negative Sampling.

The research discovered that selectively removing words led to improvements, in efficiency. By eliminating terms based on how they appear the scientists managed to accelerate the training process by 2 to 10 times. This approach of subsampling not only contributed to learning outcomes but also improved the representations of less common words. Additionally, utilizing Negative Sampling, a variation of Noise Contrastive Estimation simplified the training procedure compared to Hierarchical Softmax. Negative Sampling has proven to be a method for training models and enhancing word representations for frequently used terms.

The Skip Gram model stood out for its exceptional word representations. The vectors not only captured similarities between words but also showed a linear structure making vector arithmetic meaningful. For instance, the authors illustrated that subtracting the vector of "Spain" from "Madrid" and adding the vector of "France" resulted in a vector to "Paris". This characteristic went beyond word analogies allowing for the representation of phrases through the combination of individual word vectors. By treating occurring words as phrases and directly learning their representations the Skip Gram model effectively dealt with idiomatic expressions and grasped their meanings.

4.1 Conclusions

The key advancements in the paper focus on the introduction of Negative Sampling and the way phrases are represented. Negative Sampling offers an approach compared to Hierarchical SoftMax resulting in quicker training and improved vector representations for common words. Another notable contribution is the ability to learn meaningful phrase representations by treating them as entities. By combining these methods with subsampling words, the authors successfully trained the Skip-gram model on datasets than before resulting in high-quality word and phrase representations.

When this paper was published the techniques it presented marked a progress from existing approaches to learning word representations. The Skip-gram model's capacity to learn effectively from amounts of text data and capture semantic and syntactic relationships between words was groundbreaking. The model's performance on tasks involving reasoning and its capability to generate meaningful phrase representations set a benchmark in this field.

The influence of this research goes beyond the tasks and data sets discussed in the paper. The Skip Gram model and its related methods have been widely served as a cornerstone for subsequent research endeavors in natural language processing. By releasing the word2vec toolkit as a source that implements the Skip-gram model these effective word representations have become accessible to both researchers and practitioners driving progress in applications like text categorization, sentiment analysis, and machine translation.

In conclusion, the study on distributed word and phrase representations discussed techniques that significantly pushed forward the frontier of developing word embeddings. The Skip Gram model's effectiveness, scalability, and knack for detecting patterns have had a lasting impact on the domain opening up avenues, for exploration and real-world use in language processing.